BRAD GERVIN D. ESLABRA

Computer Engineering Graduate specializing in Embedded Systems Design

Address: Anonang, San Fabian, Pangasinan
Phone: +63 993 476 9909
Email: braddeleoneslabra@gmail.com
Website: https://capybrad.github.io/CapyBrad_Portfolio

SKILLS AND TECHNICAL PROFICIENCIES

Domain Specializations: Embedded Systems Design, Internet of Things, Firmware Programming, PCB Design and Fabrication, Software Integration and Testing
Hardware & Microcontrollers: ESP32, STM32, Arduino, Raspberry Pi
Communication Protocols: I2C, SPI, UART, MQTT
Programming Languages: C, C++, C#, Python
Developer Tools: EasyEDA, KiCAD, Microsoft Visual Studio/Visual Studio Code, Arduino IDE, STM32Cube IDE, AutoCAD, Solidworks, Git

WORK EXPERIENCE

R&D Intern - Layad Circuits Electronics Engineering Supplies and Services **May 2025 - June 2025**

- Designed and fabricated a custom ATmega-based microcontroller board, validating functionality with a bench power supply, multimeter measurements, and custom test code.
- Reduced manual serial-terminal testing of ESP boards by developing a C# based AT Command application.
- Reduced manual testing of LDR and flame sensors by designing and fabricating a test module that uses controlled LED luminosity and IR intensity to simulate light and flame conditions.
- Performed firmware verification for fellow interns' projects and unit testing for products sold by the company.

PROJECTS

LUMEN: An Integrated Sensor and Computer Vision-Based System for Rural Blind Curves Lane Safety Enhancement and Accident Detection (Capstone Project) | *GitHub Link*

- Developed the schematic and PCB Design for the system using EasyEDA.
- Trained a custom YOLOv8n computer vision model using Roboflow and Google Colab, successfully deploying and testing it on a Raspberry Pi 5 edge device.
- Developed a monitoring dashboard based on Avalonia UI Framework utilizing Tailscale and MQTT protocols for real-time telemetry communication with the deployed Raspberry Pi 5 system.

Arduino Training Kit

- Worked with a team to design a modular hardware learning kit equipped with fundamental sensors, modules, and documentation to streamline practical engineering education.
- Led the overall design and PCB fabrication of the module.

Adjustable Linear Power Supply

- Collaborated with a team to design and assemble a variable DC power supply unit based on the LM317T linear voltage regulator.
- Managed material sourcing, component selection, and the bill of materials (BOM).

EDUCATION

PHINMA University of Pangasinan **August 2022 - March 2026**

- Bachelor of Science in Computer Engineering (GWA: 1.68)

CERTIFICATIONS AND LEADERSHIP

Career Service Professional Eligibility **October 2026**

- Passed the August 2026 Career Service Examination (Professional) with a rating of 86.58%.

Editor / Executive Unit Member (CEA Student Council) **June 2025 - June 2026**

- Received the Patriotism Award for exceptional leadership and civic dedication.
- Directed publication layouts, media workflows, and technical support for council operations.